

OFFICE OF THE DEPUTY DIRECTOR

ENVIRONMENTAL HEALTH SUB-DIRECTORATE

The Air Quality Officer
Public Health Directorate
Environmental Health Sub-Directorate
P O Box 11
Port Elizabeth
6000

Tel: (041) 506 5216
Fax: (041) 506 1596
Email: airpoll@mandelametro.gov.za

**APPLICATION FORM FOR ATMOSPHERIC EMISSION LICENCE / PROVISIONAL ATMOSPHERIC
EMISSION LICENCE IN TERMS OF CHAPTER 5 OF THE NATIONAL ENVIRONMENTAL
MANAGEMENT: AIR QUALITY ACT, 2004 (ACT NO. 39 OF 2004)**

(For official use only)

File Reference Number	
Date Received	

KINDLY NOTE THAT:

- **IN THE CASE OF A NEW APPLICATION PART A AND C MUST BE FILLED IN AND SUBMITTED TO THE AEL LICENSING AUTHORITY WHEN APPLICATION FOR ENVIRONMENTAL AUTHORISATION IS MADE TO THE PROVINCIAL EIA COMPETENT AUTHORITY.**
- **PART B MUST BE FILLED IN AND SUBMITTED TO THE AEL LICENSING AUTHORITY ON COMPLETION OF THE AIR QUALITY IMPACT ASSESSMENT STUDY AND ON SUBMISSION OF THE DRAFT EIA REPORT TO THE EIA COMPETENT AUTHORITY.**
- **IN ALL OTHER APPLICATION TYPES PARTS A, B AND C MUST BE FILLED IN.**
- **IT IS THE RESPONSIBILITY OF THE APPLICANT TO ASCERTAIN WHETHER SUBSEQUENT VERSIONS OF THE FORM HAVE BEEN PRODUCED BY THE AEL LICENSING AUTHORITY.**
- **PLEASE COMPLETE ALL SECTIONS, MARK WITH AN X IN SPACES WHERE APPLICABLE.**
- **THE APPLICATION MUST BE TYPED WITHIN THE SPACES PROVIDED IN THE FORM. THE SIZES OF THE SPACES PROVIDED ARE NOT NECESSARILY INDICATIVE OF THE AMOUNT OF INFORMATION TO BE PROVIDED. SPACES ARE PROVIDED IN TABULAR FORMAT AND WILL EXTEND AUTOMATICALLY WHEN EACH SPACE IS FILLED WITH TYPING.**
- **BLACK OUT THE BOXES THAT ARE NOT APPLICABLE IN THE FORM.**
- **INCOMPLETE APPLICATIONS MAY BE RETURNED TO THE APPLICANT FOR REVISION.**
- **THE USE OF THE PHRASE "NOT APPLICABLE" IN THE FORM MUST BE DONE WITH CIRCUMSPECTION. SHOULD IT BE DONE IN RESPECT OF MATERIAL INFORMATION REQUIRED BY THE LICENSING AUTHORITY FOR ASSESSING THE APPLICATION, IT MAY RESULT IN THE REJECTION OF THE APPLICATION.**
- **FOUR (4) HARD COPIES AND ONE (1) ELECTRONIC COPY ON A COMPACT DISC OF THE APPLICATION MUST BE SUBMITTED TO THE LICENSING AUTHORITY FOR CONSIDERATION AND PROCESSING**
- **NO FAXED AND/OR E-MAILED APPLICATIONS WILL BE ACCEPTED.**
- **GRAPHICAL PROCESS INFORMATION MUST BE CLEAR, LABELED AND, WHERE APPLICABLE, SHOULD INCLUDE A TRUE NORTH ARROW AND SCALE.**

PART A

1. TYPE OF APPLICATION

	New Application		Transfer		Section 22A
	Renewal		Review	X	Variation

Current APPA Permit / Atmospheric Emission Licence Number (if applicable, please attach copy):	19/2/9/2/1/2/ NMBM AEL 13/36 (02)
--	-----------------------------------

2. ENTERPRISE INFORMATION

Entity Name	Auto X – P.E Manufacturing Facility
Trading Name (if any)	Auto-X (Pty) Ltd
Type of Entity, e.g. Company/Close Corporation/Trust, etc	Company
Company/Close Corporation/Trust Registration Number (Registration Numbers if Joint Venture)	2016/475012/07
VAT Registration Number	4220275806
Registered Address	1 Cnr Lindsay and Standford Road, Neave Township, Korsten, Port Elizabeth
Postal Address	P.O. Box 1844, Port Elizabeth, 6000
Telephone Number (General)	041 401 4000
Fax Number (General)	041 401 4000
Industry Type/Nature of Trade	Lead Acid Battery Manufacturing
SMME Status (Total gross asset value, excluding fixed property in Rand)	R 1 368 333 496.00
Land Use Zoning as per Town Planning Scheme	Industrial
Land Use Rights if outside Town Planning Scheme	N/A

Name of Contact Person	Tshepo Nkolanyane
Telephone Number	041 401 4040
Cell Phone Number	073 262 0355
Fax Number	041 401 4000
E-mail Address	TshepoNK@auto-x.co.za
After Hours Contact Details	073 262 0355

3. LISTED ACTIVITIES TO BE CONDUCTED:

List all Listed Activities, as published in terms of section 21 of the National Environmental Management: Air Quality Act, 2004 (Act No. 39 of 2004), proposed to be conducted at the premises in terms of this application:

Category of Listed Activity	Sub-Category of Listed Activity	Listed Activity Name	Description of the Listed Activity
4	4.13	Lead Smelting	The processing of lead by the application of heat and the production of electric batteries containing lead.
7	7.2	Production of acids	The production, bulk handling and or use in manufacturing of hydrofluoric, hydrochloric, nitric and sulphuric acid (including oleum) in concentration exceeding 10%. Processes in which oxides of sulphur are emitted through the production of acid sulphites of alkalis or alkaline earths or through the production of liquid sulphur or sulphurous acid. Secondary production of hydrochloric acid through regeneration.

Despite the repeal of the Atmospheric Pollution Prevention Act, 1965 (Act No. 45 of 1965), list all Scheduled Process(es), as was or were set out in the Second Schedule of the repealed Atmospheric Pollution Prevention Act, 1965, currently conducted at the premises:

APPA Registration Certificate Number	Date of Registration Certificate	Scheduled Process Number	Scheduled Process Description

4. LOCATION AND EXTENT OF PLANT

Physical Address of the Licensed Premises	1 Cnr Lindsay and Standford Road, Neave Township, Korsten, Port Elizabeth
Description of Site (Erf)	2619 & 2620
Coordinates (latitude, longitude) of Approximate Center of Operations (Decimal Degrees)	Latitude: -33.9267 Longitude: 25.5724
Coordinates of Approximate Center of Operations	North-south: 25° 34' 20.6394" East-west: -33° 55' 36.1194"
Extent (km ²)	0.055

Elevation Above Mean Sea Level (m)	46
Province	Eastern Cape
Metropolitan Municipality	Nelson Mandela Bay Municipality
Designated Priority Area (if applicable)	N/A

5. DESCRIPTION OF SURROUNDING LAND USE (within 5 km radius)

Provide a description of the surrounding land use within a 5 km radius, specifically noting the names and proximity of residential and commercial areas in relation to the site of the works.

North	Approximately 0.9 km north of the AutoX facility is the Young Park residential area that extends to 1.6 km north of the facility. Algoa Park extends from 1.9 km north of the facility to approximately 4.7 km north of AutoX. Salt Pan extends from 4.7 km north of the site to 5 km north of the site. Additionally, the Kwaford, Kwazakhele and New Brighton residential areas are situated in a northerly direction from the AutoX facility.
East	Neave Industrial extends to 0.8 km east of the site. 1.3 km east of AutoX lies Holland Park, which is bordered by Ferguson and Sydenham residential areas at approximately 1.5 km and 2 km east, respectively, of the site. The Indian Ocean is situated 3.5 km east of the facility, while the North End industrial area lies 2.7 km southeast of AutoX.
South	Korsten residential area extends for 1.2 km south of AutoX. Approximately 1.2 km south of AutoX lies the Newton Park residential area. Newton Park extends to 4 km south of the facility, and 4.8 km south of the site lies the Walmer residential area.
West	West of the facility lies the Schauderville residential area, which extends for 1.4 km west of AutoX. The Parkside residential area is approximately 1.6 km west of the site and extends for 2.2 km west of AutoX. The Malabar residential area is 2.5 km west of AutoX, which is met by the Westering residential area approximately 3.6 km west of the site.

Attach map(s), satellite image(s) or aerial photograph(s) detailing location of premises in relation to surrounding community.



PART B

6. NATURE OF PROCESS

6.1 Process Description

Please provide a detailed description of the entire production process including reference to the overall balance sheet of inputs, outputs and emissions at the site of the works.

1) Lead melting and Lead Oxide Manufacturing:

- Lead ingots are fed via a conveyor to a melting pot.
- Pure molten lead from the melting pot is fed into a reactor along with water and ambient air.
- In the reactor, the molten lead is oxidized to form lead oxide (PbO , PbO_2 , Pb_2O_3).
- The lead oxide particles are blown up through a classifier, which separates the large particles from the small particles. The large particles fall down through the classifier back into the reactor.
- Fine lead oxide particles exit the classifier and travel to a cyclone, which is followed by a bag filter.
- After the bag filter, a high-efficiency particulate arrestor (HEPA) filter separates the lead oxide particles from the air stream.
- Air that is removed from the stream exits the exhaust through the post-classifier stacks AX14 and AX15.
- Air is introduced at the blending screw conveyor to transport the lead oxide through the feed line to the storage silos.
- The transport air is then cleaned via a bag filter and exhausted through stacks AX12 and AX13.
- The lead oxide manufacturing area is served by a multi-point extraction system that feeds into stack AX10.

2) Plate Manufacturing (Positive and Negative):

- In the plate strip casting process, lead ingots are melted down and cast to create lead strip coils.
- In the Plate Manufacturing Process, the battery plates are made from the lead strip coils produced in the Plate Strip Casting process.
- A lead strip coil is fed through the expander line to expand the strip into a parallel pair of continuous grids that are fed through the paster, where they are coated with Lead Oxide paste.
- The coated grids are then cut to size to form the positive and negative plates.
- In the Paste Mixing process, ambient air, water, acid, and other additives are mixed with lead oxide to form a paste.
- The paste mixer is served by a wet scrubber that leads into stack AX16.
- The plates are placed into curing ovens for processes, after which they are stored in the staging area for work in progress.
- Stack AX9 is an extraction stack for activities within the plate manufacturing process,

including the molten lead pots that may release lead fumes.

- There are two types of curing ovens present at AutoX: indirectly heat gas-fired ovens and electric ovens.
- Curing occurs between 40°C and 50°C, allowing crystal lattices to form in the paste.

3) Battery Machine (BM) Assembly and TBS:

- In the plate wrapping stage, positive plates are enveloped and interleaved with negative plates to form an element.
- The elements are held in the buffer area before they are transferred to the loading station, where the interleaved plate elements are loaded into the moulds.
- In the Cell Group Burning stage, positive and negative element lugs are welded together to form a cell.
- Cells are dropped into cases, and Through The Partition (TTP), welding takes place to weld each cell together.
- Covers/lids are heat-sealed onto the cases in the Heat-Sealing Station.
- In Post burning, the negative and positive end posts/terminals are automatically burnt before the batteries undergo a leak test. Liquid grease is applied to end post/terminals and the unfinished batteries are automatically packed into pellets.
- Extraction units serve various processes within BM assembly, which leads to stacks AX6, AX7, AX8 and AX22.
- Stack AX1 serves the TBS assembly area, while stack AX19 serves the La Pneumatic lid sealing machine.

4) Charge Room (Battery Formation and After Charging):

- In the charge room, the batteries are filled with sulphuric acid. The Battery Acid Filling machine is served by four extraction units (AX20) that vent directly outside the building.
- The batteries are then charged as per applicable charging profiles.
- The batteries are washed and rinsed in washing tunnels before they are cleaned with chemicals and packaged for shipping.
- Stacks AX2 and AX3 serve the charge room.

6.2 Unit Processes

List all unit processes associated with the listed activities in operation at the premises by the atmospheric emission licence holder, highlighting unit processes proposed in respect of this application:

Unit Process	Unit Process Function	Batch or Continuous Process
Receiving	Receiving of raw materials	Batch
Storage	Storage of lead ingots and other raw materials in warehouse	Continuous
Storage – Tank farm	Storage and dilution of sulphuric acid in tank farm	Continuous

Lead Oxide Production	Production of lead oxide from molten lead, water and air. Classification of lead oxide particles. Conveyance of lead oxide to storage silos.	Continuous
Plate Production	Lead ingots are melted and cast into strip coils. Coils are expanded to form grids. Lead oxide is combined with water, sulphuric acid and other additives to form lead oxide paste. Lead oxide paste is applied to grids to form plates. Plates are cured in ovens.	Continuous
Battery assembly	Positive plates are enveloped and interleaved with negative plates to form elements. Elements are loaded into moulds. Elements are welded together to form cells. Cells are dropped into cases and welded together. Covers/lids are heat-sealed onto the cases. End terminals are burnt. The batteries undergo a leak test. This application includes the addition of a third battery assembly line (BM Line 3), identical to existing BM Lines 1 and 2.	Continuous
Charging	Batteries are filled with sulphuric acid and charged.	Batch
Cleaning	Batteries are washed in tunnels and cleaned with chemicals before being packed for shipping.	Batch
Shipping	Batteries are dispatched.	Batch

Please provide any other unit processes currently conducted at the site of works.

Name of the Unit Process	Description of the Unit Process

6.3 Hours of Operation

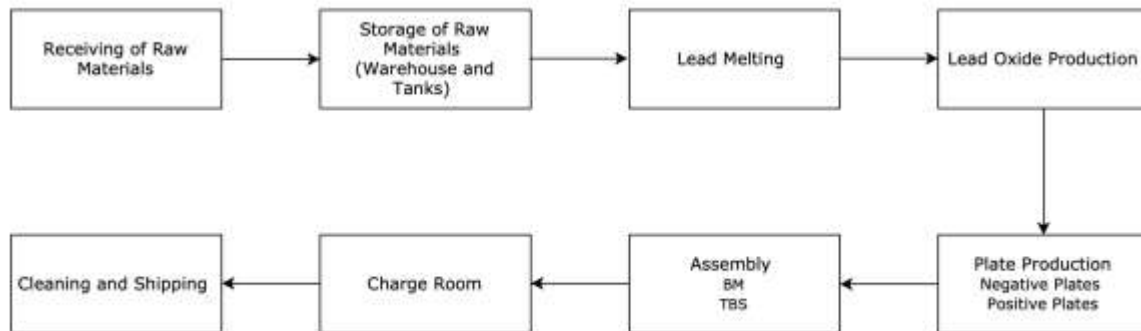
Provide the hours of operation of all unit processes associated with the listed activities in operation at the premises by the atmospheric emission licence holder, highlighting unit processes proposed in respect of this application:

Unit Process	Operating Hours	No. Days Operation per Year
Receiving	24 hrs per day/ 7 days per week	365
Storage	24 hrs per day/ 7 days per week	365
Lead Oxide Production	24 hrs per day/ 7 days per week	365
Plate Manufacturing	24 hrs per day/ 7 days per week	365
Battery Assembly	24 hrs per day/ 7 days per week	365
Charging	24 hrs per day/ 7 days per week	365
Cleaning	24 hrs per day/ 7 days per week	365
Shipping	24 hrs per day/ 7 days per week	365

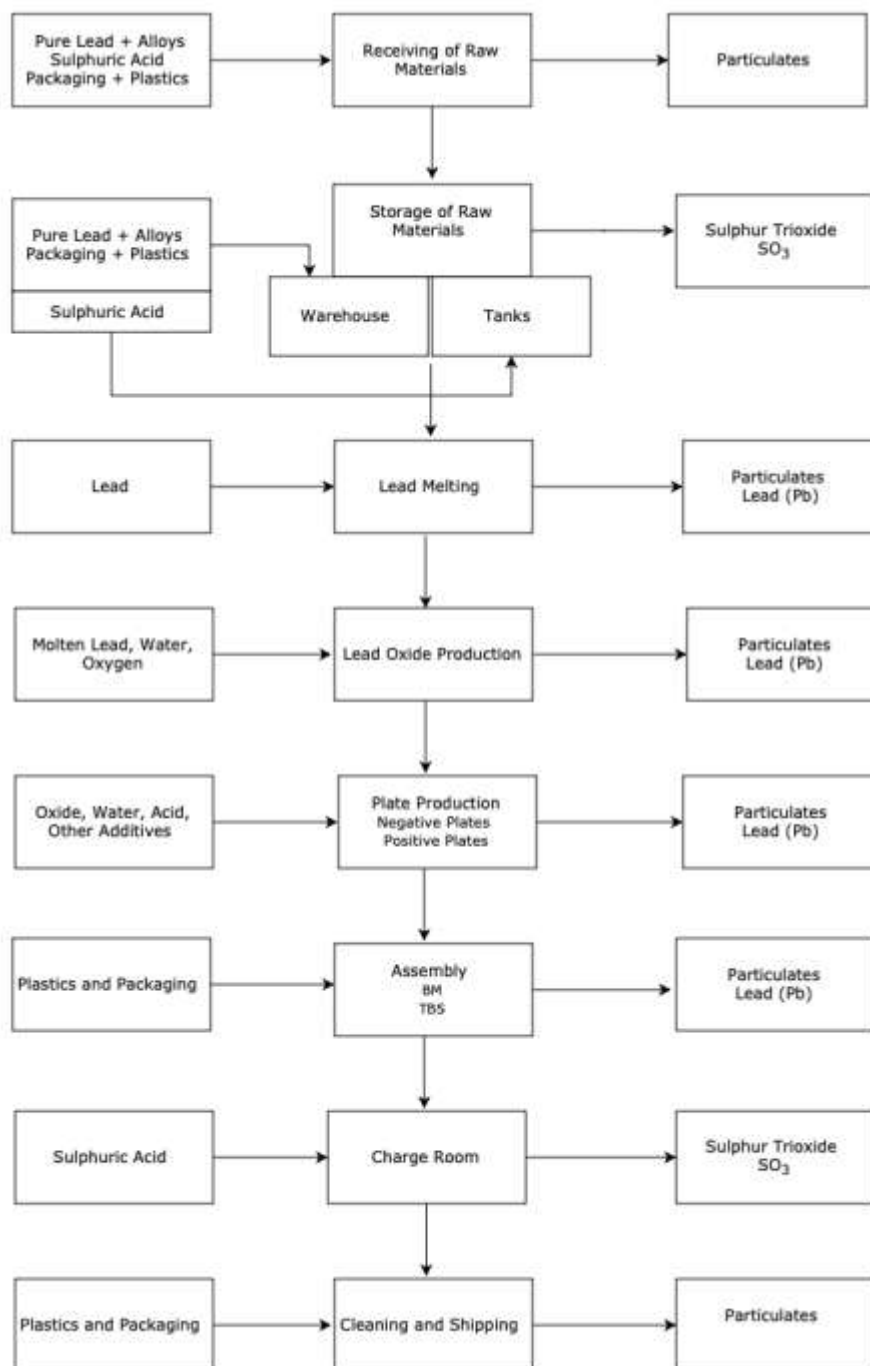
6.4 Graphical Process Information

Attach the following for the entire operation being undertaken at the site of the works:

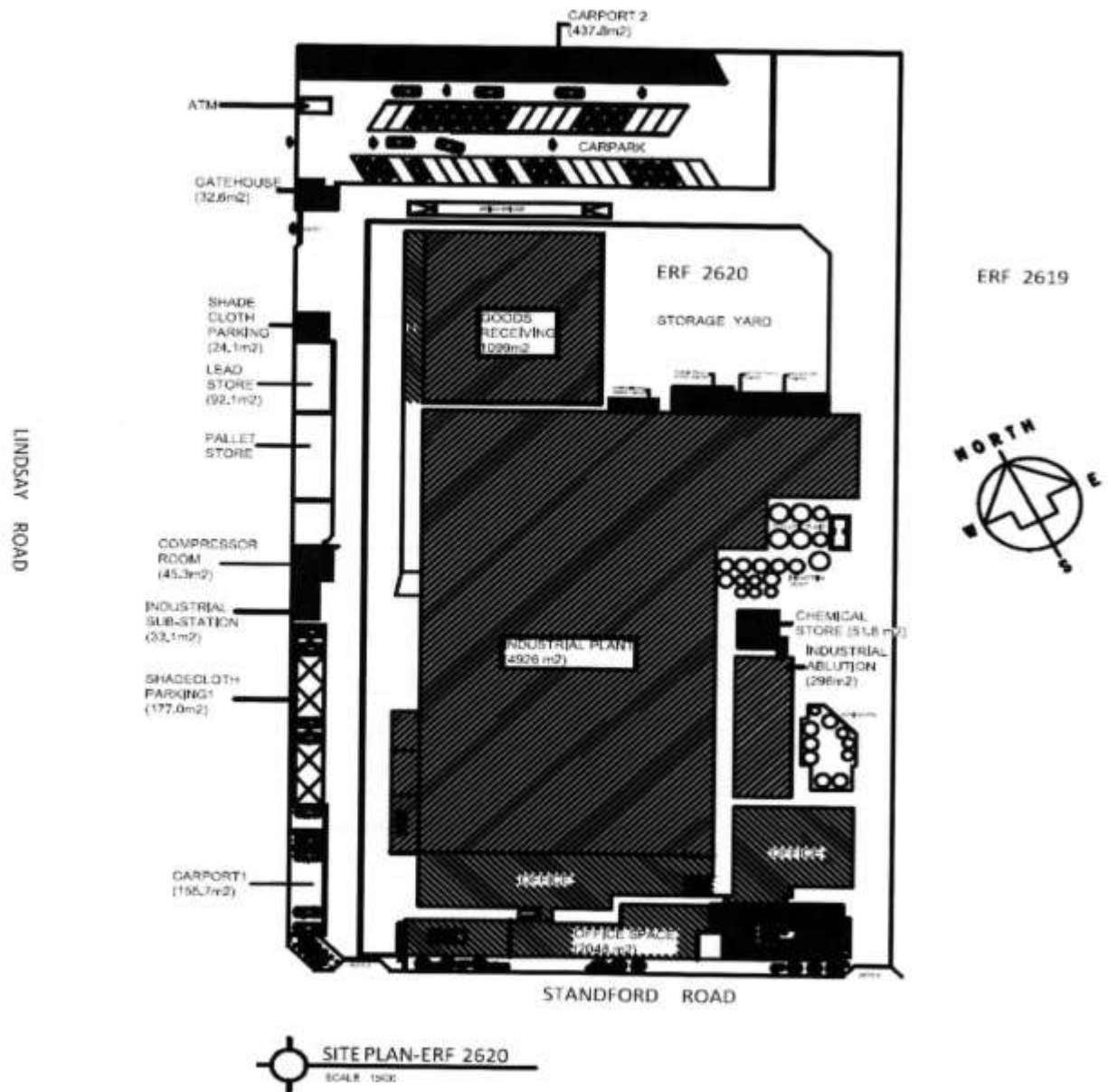
- Simplified block diagram with the name of each unit process in a block; showing links between all unit processes or blocks.



- Process flow chart(s) clearly indicating inputs, outputs and emissions at the site of works, including points of potential fugitive emissions and emergency releases.



-
- SITE PLAN - ERF 2619**
SCALE 1:1000
- ERF 2620**
- ERF 2619**
- STANDFORD ROAD**
- GATE 4**
- GATE 5**
- PLATE MANUFACTURING**
(4779.1 m²)
- DC COMPONENT 2**
(1065.8 m²)
- SAGERY SUB-STATION 4**
(104.2 m²)
- DC COMPONENT 2 SUB-STATION 4**
(104.2 m²)
- CHILL WATER TOWERS**
(102.6 m²)
- TECH STORES**
(719.2 m²)
- HIGH WORKSHOP**
(503.9 m²)
- GYMNASIUM**
(1105.6 m²)
- POSSIBLE TECH OFFICES**
(357.0 m²)
- CHANGING AREA**
(4200.9 m²)
- LOGISTICS AREA STORAGE**
(751.8 m²)
- STC**
(1398.3 m²)
- ERF 2619**
- ERF 2620**
- STANDFORD ROAD**
- GATE 4**
- GATE 5**
- PLATE MANUFACTURING**
(4779.1 m²)
- DC COMPONENT 2**
(1065.8 m²)
- SAGERY SUB-STATION 4**
(104.2 m²)
- DC COMPONENT 2 SUB-STATION 4**
(104.2 m²)
- CHILL WATER TOWERS**
(102.6 m²)
- TECH STORES**
(719.2 m²)
- HIGH WORKSHOP**
(503.9 m²)
- GYMNASIUM**
(1105.6 m²)
- POSSIBLE TECH OFFICES**
(357.0 m²)
- CHANGING AREA**
(4200.9 m²)
- LOGISTICS AREA STORAGE**
(751.8 m²)
- STC**
(1398.3 m²)
- ERF 2619**
- ERF 2620**
- STANDFORD ROAD**
- GATE 4**
- GATE 5**



NB: Indicate clearly on the above graphics the listed activity or activities applied for in this application. Alternatively, provide additional graphics for the listed activity or activities applied for.

7 RAW MATERIALS AND PRODUCTS

Provide production and by-production rates, raw material information and emissions information.

7.1 Raw Materials Used

Raw Material Type	Maximum Permitted Consumption Rate (Volume)	Design Consumption Rate (Volume)	Actual Consumption Rate (Volume)	Units (Quantity/Period)
Pure Lead and Alloys	25 998 910	25 998 910	17 937 186	kg/annum
Plastic Components (cases, covers, vents, etc.)	16 670 790	16 670 790	10 518 013	QTY/annum
Labels	41 968 500	41 968 500	5 450 133	Units/annum
Sulphuric Acid (diluted)	9 734 192	9 734 192	5 616 407	Litres/annum
Separators	5 504 067	5 504 067	2 483 067	m ² /annum
Polyester Paper	16 000	16 000	15 492	Rolls/annum
Premaflok Fibre	43 162	43 162	11 975	kg/annum
TBLS Plus	122 424	122 424	77 000	kg/annum
PBX Carbon	5 500	5 500	5 295	kg/annum
Expenders	75 186	75 186	43 000	kg/annum
Caustic	270 178	270 178	149 100	kg/annum

7.2 Production Rates

Production Name	Maximum Production Capacity Permitted (Volume)	Design Production Capacity (Volume)	Actual Production Capacity (Volume)	Units (Quantity/Period)
Automotive Batteries	2 533 450	2 533 450	1 599 647	Units/annum

By-Product Name	Maximum Production Capacity Permitted (Volume)	Design Production Capacity (Volume)	Actual Production Capacity (Volume)	Units (Quantity/Period)

--	--	--	--	--

7.3 Energy Sources Used

Energy Source	Sulphur Content of Fuel (%) (if applicable)	Ash Content of Fuel (%) (if applicable)	Maximum Permitted Consumption Rate (Volume)	Design Consumption Rate (Volume)	Actual Consumption Rate (Volume)	Units (Quantity/Period)
Electricity	N/A	N/A	2 200 MWh	2 200 MWh	2 050 MWh	MWh/month
LPG	0	0	250 tonnes/annum	250 tonnes/annum	185 tonnes/annum	Tonnes/annum

7.4 Sources of Atmospheric Emission (including greenhouse gas emissions)

7.4.1 Point Source Parameters

Unique Stack ID	Source Name	Latitude (decimal degrees)	Longitude (decimal degrees)	Height of Release Above Ground (m)	Height Above Nearby Building (m)	Diameter at Stack Tip / Vent Exit (m)	Actual Gas Exit Temperature (°C)	Actual Gas Volumetric Flow (m³/hr)	Actual Gas Exit Velocity (m/s)
AX1	TBS Assembly	-33.92619001	25.57375359	4	0	0.9	19.97	24 972	10.90
AX2	Charge Room Extraction Stack 1	-33.92645603	25.57354351	8	0	1.2	29.59	63 120	15.50
AX3	Charge Room Extraction Stack 2	-33.92715606	25.57294874	8	0	1.2	36.36	67 568	16.60
AX4	BM Assembly Extraction 4	-33.926294	25.574358	6	0	1	30.0	25 415	9.0
AX6	BM Assembly Extraction 1	-33.92635657	25.57445474	6	1	1	23.90	44 630	12.40
AX7	BM Assembly Extraction 2	-33.92643524	25.57459071	6	1	1	25.93	44 751	12.43
AX8	BM Assembly Extraction 3	-33.92650929	25.57470794	6	1	1	25.25	31 109	8.64
AX9	Plate Making Extraction	-33.92708845	25.57564552	5	0	0.9	36.78	71 442	31.19

Unique Stack ID	Source Name	Latitude (decimal degrees)	Longitude (decimal degrees)	Height of Release Above Ground (m)	Height Above Nearby Building (m)	Diameter at Stack Tip / Vent Exit (m)	Actual Gas Exit Temperature (°C)	Actual Gas Volumetric Flow (m³/hr)	Actual Gas Exit Velocity (m/s)
AX10	Lead Oxide Manufacturing Extraction	-33.92738291	25.57510967	6.5	0	1.2	31.12	23 795	5.84
AX12	Oxide Mill Silo Stack (Negative Oxide)	-33.92741364	25.57539616	5.5	0	0.4	74.24	3 361	7.43
AX13	Oxide Mill Silo Stack (Positive Oxide)	-33.92745583	25.57534883	5.5	0	0.5	98.84	3 757	5.32
AX14	Post-Classifer Bag Filters 1, 2 and 3	-33.92656138	25.57481471	8.5	1	0.25	32.22	1 259	7.12
AX15	Post-Classifer Bag Filters 4,5 and 6	-33.92725642	25.57540916	7.75	1	0.25	46.46	2 181	12.34
AX16	Paste Mixer Stack	-33.92723596	25.57537110	6	1	0.5 x 0.45	25.47	2 564	3.17
AX17A	Curing Oven Electric	-33.926896	25.574992	6.5	1	0.3 x 0.3	49.71	4 777	14.74
AX17B	Curing Oven Electric	-33.927113	25.574962	6.5	1	0.3 x 0.3	49.71	4 777	14.74

Unique Stack ID	Source Name	Latitude (decimal degrees)	Longitude (decimal degrees)	Height of Release Above Ground (m)	Height Above Nearby Building (m)	Diameter at Stack Tip / Vent Exit (m)	Actual Gas Exit Temperature (°C)	Actual Gas Volumetric Flow (m³/hr)	Actual Gas Exit Velocity (m/s)
AX17C	Curing Oven Electric	-33.92714	25.574937	6.5	1	0.3 x 0.3	49.71	4 777	14.74
AX17D	Curing Oven Electric	-33.927164	25.574916	6.5	1	0.3 x 0.3	49.71	4 777	14.74
AX17E	Curing Oven Electric	-33.92719	25.574894	6.5	1	0.3 x 0.3	49.71	4 777	14.74
AX17F	Curing Oven Electric	-33.92694	25.574834	6.5	1	0.3 x 0.3	49.71	4 777	14.74
AX18A	Curing Oven Gas Fired	-33.926771	25.574975	7	1	0.35	46.35	614	1.77
AX18B	Curing Oven Gas Fired	-33.926792	25.574956	7	1	0.35	46.35	614	1.77
AX18C	Curing Oven Gas Fired	-33.926809	25.574941	7	1	0.35	46.35	614	1.77
AX18D	Curing Oven Gas Fired	-33.926829	25.574924	7	1	0.35	46.35	614	1.77
AX18E	Curing Oven Gas Fired	-33.926845	25.57491	7	1	0.35	46.35	614	1.77
AX18F	Curing Oven Gas Fired	-33.926866	25.574891	7	1	0.35	46.35	614	1.77
AX18G	Curing Oven Gas Fired	-33.926884	25.574874	7	1	0.35	46.35	614	1.77

Unique Stack ID	Source Name	Latitude (decimal degrees)	Longitude (decimal degrees)	Height of Release Above Ground (m)	Height Above Nearby Building (m)	Diameter at Stack Tip / Vent Exit (m)	Actual Gas Exit Temperature (°C)	Actual Gas Volumetric Flow (m³/hr)	Actual Gas Exit Velocity (m/s)
AX18H	Curing Oven Gas Fired	-33.926906	25.574856	7	1	0.35	46.35	614	1.77
AX19	La Pneumatic Stack (Lid Sealing)	-33.92679250	25.57495557	7.5	1	0.4	34.56	1 543	3.41
AX20A	Battery Acid Filling	-33.92651953	25.57387954	5	0	0.45	22.89	5 895	10.30
AX20B	Battery Acid Filling	-33.92651953	25.57387954	5	0	0.45	22.89	5 895	10.30
AX20C	Battery Acid Filling	-33.92651953	25.57387954	5	0	0.45	22.89	5 895	10.30
AX20D	Battery Acid Filling	-33.92651953	25.57387954	5	0	0.45	22.89	5 895	10.30
AX22	BM Assembly Extraction 4	-33.92661812	25.57360138	4	0	0.25	25.07	4 427	25.05
AX23	Proposed Expansion Stack 1	-33.92665201	25.57356837	6	2	0.55	55.1	11 784	13.78
AX24	Proposed Expansion Stack 2	-33.92666048	25.57367641	4	0	0.55	32.9	18 834	22.02
AX25	Proposed Expansion Stack 3	-33.92670686	25.57363236	6	2	0.25	25.07	4 427	25.05

Unique Stack ID	Source Name	Latitude (decimal degrees)	Longitude (decimal degrees)	Height of Release Above Ground (m)	Height Above Nearby Building (m)	Diameter at Stack Tip / Vent Exit (m)	Actual Gas Exit Temperature (°C)	Actual Gas Volumetric Flow (m³/hr)	Actual Gas Exit Velocity (m/s)
AX26	Proposed Expansion Stack 4	-33.92643965	25.57240269	6	2	0.25	25.07	4 427	25.05
AX27	Proposed Expansion Stack 5	-33.92640449	25.57237970	6	2	0.25	25.07	4 427	25.05
AX28	Proposed Expansion Oven Stack	-33.9263981	25.5724342	6	2	0.25	122.00	1 044	5.91

7.4.2 Point Source Emissions

As per 5.4.1 ID Code	Pollutant Name	Maximum Hourly Release Rate			Maximum Daily Release Rate			Average Annual Release Rate			Emissions Hours (e.g. 07h00 – 17h00)	Type of Emissions (Continuous / Routine but Intermittent / Emergency Only)
		(mg/Nm ³)	(mg/Am ³)	(g/s)	(mg/Nm ³)	(mg/Am ³)	(tons/day)	(mg/Nm ³)	(mg/Am ³)	(tpa)		
AX1	Particulate Matter	30		0.18	30		0.016	3		0.67	00h00 – 12h00	Continuous
	Lead (Pb)	2		0.012	2		0.001	0.27		0.05	00h00 – 12h00	
AX2	Sulphur Trioxide (SO ₃)	25		25	25		2.16	0.06		0.028	00h00 – 12h00	Continuous
	Sulphur Dioxide (SO ₂)	350		5.524	350		0.48	3		1.49	00h00 – 12h00	
	Oxides of Nitrogen	350		5.524	350		0.48	0		0	00h00 – 12h00	
AX3	Sulphur Trioxide (SO ₃)	25		0.413	25		0.035	0.05		0.025	00h00 – 12h00	Continuous
	Sulphur Dioxide (SO ₂)	350		5.78	350		0.5	4		2.08	00h00 – 12h00	
	Oxides of Nitrogen	350		5.78	350		0.5	3		1.56	00h00 – 12h00	

As per 5.4.1 ID Code	Pollutant Name	Maximum Hourly Release Rate			Maximum Daily Release Rate			Average Annual Release Rate			Emissions Hours (e.g. 07h00 – 17h00)	Type of Emissions (Continuous / Routine but Intermittent / Emergency Only)
		(mg/Nm ³)	(mg/Am ³)	(g/s)	(mg/Nm ³)	(mg/Am ³)	(tons/day)	(mg/Nm ³)	(mg/Am ³)	(tpa)		
AX4	Particulate Matter	30		0.3297	30		0.0183	2		6.68	00h00 – 12h00	Continuous
	Lead (Pb)	2		0.022	2		0.00122	0.13		0.445	00h00 – 12h00	
AX6	Particulate Matter	30		0.3297	30		0.028	2		3.26	00h00 – 12h00	Continuous
	Lead (Pb)	2		0.022	2		0.002	0.13		0.044	00h00 – 12h00	
AX7	Particulate Matter	30		0.326	30		0.028	2		0.63	00h00 – 12h00	Continuous
	Lead (Pb)	2		0.022	2		0.002	0.03		0.012	00h00 – 12h00	
AX8	Particulate Matter	30		0.2256	30		0.019	14		3.26	00h00 – 12h00	Continuous
	Lead (Pb)	2		0.015	2		0.0012	0.08		0.018	00h00 – 12h00	
AX9	Particulate Matter	30		0.5205	30		0.04	5		2.48	00h00 – 12h00	Continuous
	Lead (Pb)	2		0.0347	2		0.003	0.05		0.033	00h00 – 12h00	
AX10	Particulate Matter	30		0.1731	30		0.015	3		0.601	00h00 – 12h00	Continuous

As per 5.4.1 ID Code	Pollutant Name	Maximum Hourly Release Rate			Maximum Daily Release Rate			Average Annual Release Rate			Emissions Hours (e.g. 07h00 – 17h00)	Type of Emissions (Continuous / Routine but Intermittent / Emergency Only)
		(mg/Nm ³)	(mg/Am ³)	(g/s)	(mg/Nm ³)	(mg/Am ³)	(tons/day)	(mg/Nm ³)	(mg/Am ³)	(tpa)		
	Lead (Pb)	2		0.0115	2		0.001	0.16		0.03	00h00 – 12h00	
AX12	Particulate Matter	30		0.0207	30		0.002	3		0.059	00h00 – 12h00	Continuous
	Lead (Pb)	2		0.0014	2		0.0001	0.78		0.017	00h00 – 12h00	
AX13	Particulate Matter	30		0.0202	30		0.0017	4		0.076	00h00 – 12h00	Continuous
	Lead (Pb)	2		0.0013	2		0.0001	0,69		0.015	00h00 – 12h00	
AX14	Particulate Matter	30		0.0092	30		0.001	3		0.031	00h00 – 12h00	Continuous
	Lead (Pb)	2		0.0006	2		0	1,26		0.012	00h00 – 12h00	
AX15	Particulate Matter	30		0.0153	30		0.001	6		0.108	00h00 – 12h00	Continuous
	Lead (Pb)	2		0.0010	2		0.0001	0.25		0.0047	00h00 – 12h00	
AX16	Particulate Matter	30		0.019	30		0.0013	5		0.109	00h00 – 12h00	Continuous
	Lead (Pb)	2		0.0013	2		0.0001	0.7		0.0047	00h00 – 12h00	

As per 5.4.1 ID Code	Pollutant Name	Maximum Hourly Release Rate			Maximum Daily Release Rate			Average Annual Release Rate			Emissions Hours (e.g. 07h00 – 17h00)	Type of Emissions (Continuous / Routine but Intermittent / Emergency Only)
		(mg/Nm ³)	(mg/Am ³)	(g/s)	(mg/Nm ³)	(mg/Am ³)	(tons/day)	(mg/Nm ³)	(mg/Am ³)	(tpa)		
AX17	Particulate Matter	30		0.0291	30		0.0025	3		0.074	00h00 – 12h00	Continuous
	Lead (Pb)	2		0.0019	2		0.00016	0.22		0.005	00h00 – 12h00	
AX18	Particulate Matter	30		0.0041	30		0.00035	5		0.0218	00h00 – 12h00	Continuous
	Lead (Pb)	2		0.0003	2		0.00003	0.21		0.0009	00h00 – 12h00	
AX19	Particulate Matter	30		0.0111	30		0.001	8		0.094	00h00 – 12h00	Continuous
	Lead (Pb)	2		0.0007	2		0	0.89		0.01	00h00 – 12h00	
AX20	Sulphur Trioxide (SO ₃)	25		0.0375	25		0.0032	0.05		0.0032	00h00 – 12h00	Continuous
	Sulphur Dioxide (SO ₂)	350		0.5252	350		0.045	11		0.52	00h00 – 12h00	
	Oxides of Nitrogen	350		0.5252	350		0.045	2		0.095	00h00 – 12h00	
AX22	Particulate Matter	30		0.0333	30		0.0029	1		0.038	00h00 – 12h00	Continuous

As per 5.4.1 ID Code	Pollutant Name	Maximum Hourly Release Rate			Maximum Daily Release Rate			Average Annual Release Rate			Emissions Hours (e.g. 07h00 – 17h00)	Type of Emissions (Continuous / Routine but Intermittent / Emergency Only)
		(mg/Nm ³)	(mg/Am ³)	(g/s)	(mg/Nm ³)	(mg/Am ³)	(tons/day)	(mg/Nm ³)	(mg/Am ³)	(tpa)		
	Lead (Pb)	2		0.0022	2		0.00019	0.33		0.012	00h00 – 12h00	
AX23	Particulate Matter	30		0.081	30		0.007	4.75		0.404	00h00 – 12h00	Continuous
	Lead (Pb)	2		0.0054	2		0.0005	0.1425		0.0126	00h00 – 12h00	
AX24	Particulate Matter	30		0.1365	30		0.012	4.75		0.681	00h00 – 12h00	Continuous
	Lead (Pb)	2		0.0091	2		0.0008	0.1425		0.019	00h00 – 12h00	
AX25	Particulate Matter	30		0.0333	30		0.0029	4.75		0.167	00h00 – 12h00	Continuous
	Lead (Pb)	2		0.0022	2		0.0002	0.1425		0.0063	00h00 – 12h00	
AX26	Particulate Matter	30		0.0333	30		0.0029	4.75		0.167	00h00 – 12h00	Continuous
	Lead (Pb)	2		0.0022	2		0.0002	0.1425		0.0063	00h00 – 12h00	
AX27	Particulate Matter	30		0.0333	30		0.0029	4.75		0.167	00h00 – 12h00	Continuous
	Lead (Pb)	2		0.0022	2		0.0002	0.1425		0.0063	00h00 – 12h00	

As per 5.4.1 ID Code	Pollutant Name	Maximum Hourly Release Rate			Maximum Daily Release Rate			Average Annual Release Rate			Emissions Hours (e.g. 07h00 – 17h00)	Type of Emissions (Continuous / Routine but Intermittent / Emergency Only)
		(mg/Nm ³)	(mg/Am ³)	(g/s)	(mg/Nm ³)	(mg/Am ³)	(tons/day)	(mg/Nm ³)	(mg/Am ³)	(tpa)		
AX28	Particulate Matter	50		0.0090	50		0.00078	-		0.0129	00h00 – 12h00	Continuous
	Sulphur Dioxide (SO ₂)	500		0.0896	500		0.0077	-		0	00h00 – 12h00	
	Oxides of Nitrogen	500		0.0896	500		0.0077	-		0.41	00h00 – 12h00	

7.4.3 Point Source Current Emissions Monitoring

Provide information on current/proposed emission monitoring, including requirements as set out in currently held atmospheric emission licence(s), where applicable, for existing operations at the site of works

Attach a copy of all air pollution monitoring reports conducted for the last three years, where applicable.

As per 5.4.1 ID	Emission Sampling / Monitoring Method	Sampling Frequency	Sampling Duration	Measured Parameters
AX1; 4; 6; 7; 8; 9; 10; 12; 13; 14; 15; 16; 17; 18; 19; 22; 23; 24; 25; 26; 27; 28	USEPA Method 5, 17 and 29	Annual	Minimum of 3 x 1-hour tests	Particulates and Lead
AX 2; 3; 20	USEPA Method 6c, 7e and 8	Annual	Minimum of 3 x 1-hour tests	SO ₂ , NO _x , SO ₃ / H ₂ SO ₄ Mist

7.4.4 Point Source Emission Estimation Information

As per 5.4.1 ID	Basis for Emission Rates
AX1	Assumed 24 hrs/day; 365 days a year
AX2	Assumed 24 hrs/day; 365 days a year
AX3	Assumed 24 hrs/day; 365 days a year
AX4	Assumed 24 hrs/day; 365 days a year
AX6	Assumed 24 hrs/day; 365 days a year
AX7	Assumed 24 hrs/day; 365 days a year
AX8	Assumed 24 hrs/day; 365 days a year
AX9	Assumed 24 hrs/day; 365 days a year
AX10	Assumed 24 hrs/day; 365 days a year
AX12	Assumed 24 hrs/day; 365 days a year

As per 5.4.1 ID	Basis for Emission Rates
AX13	Assumed 24 hrs/day; 365 days a year
AX14	Assumed 24 hrs/day; 365 days a year
AX15	Assumed 24 hrs/day; 365 days a year
AX16	Assumed 24 hrs/day; 365 days a year
AX17	Assumed 24 hrs/day; 365 days a year
AX18	Assumed 24 hrs/day; 365 days a year
AX19	Assumed 24 hrs/day; 365 days a year
AX20	Assumed 24 hrs/day; 365 days a year
AX22	Assumed 24 hrs/day; 365 days a year
AX23	Assumed 24 hrs/day; 365 days a year
AX24	Assumed 24 hrs/day; 365 days a year
AX25	Assumed 24 hrs/day; 365 days a year
AX26	Assumed 24 hrs/day; 365 days a year
AX27	Assumed 24 hrs/day; 365 days a year
AX28	Assumed 24 hrs/day; 365 days a year

7.4.5 Area and/or Line Source Parameters

Unique Area Source ID	Source Name	Source Description	Latitude (decimal degrees) of SW corner	Longitude (decimal degrees) of SW corner	Height of Release Above Ground (m)	Length of Area (m)	Width of Area (m)	Angle of Rotation from True North (°)
S1	Tank S1	H ₂ SO ₄ – 98%	-33.926	25.527	4.6	21	12	350
S2	Tank S2	H ₂ SO ₄ – 98%	-33.926	25.527	4.6	21	12	350
S3	Tank S3	H ₂ SO ₄ – 18%	-33.926	25.527	4.6	21	12	350
S4	Tank S4	H ₂ SO ₄ – 18%	-33.926	25.527	2.7	21	12	350
S5	Tank S5	H ₂ SO ₄ – 50%	-33.926	25.527	4.6	21	12	350
S6	Tank S6	H ₂ SO ₄ – 45%	-33.926	25.527	2.7	21	12	350
S7	Tank S7	Recovery – 35%	-33.926	25.527	2.7	21	12	350
S8	Tank S8	H ₂ SO ₄ – 50%	-33.926	25.527	4.6	21	12	350
S9	Tank S9	Recovery – 35%	-33.926	25.527	4.6	21	12	350
S10	Tank S10	H ₂ SO ₄ – 98%	-33.926	25.527	4.6	21	12	350
S11	Tank S11	H ₂ SO ₄ – 45%	-33.926	25.527	4.6	21	12	350
S12	Tank S12	H ₂ SO ₄ – 45%	-33.926	25.527	4.6	21	12	350
S13	Tank S13	H ₂ SO ₄ – 18%	-33.926	25.527	5	3	8.5	45
S14	Tank S14	H ₂ SO ₄ – 18%	-33.926	25.527	5	3	8.5	340
S15	Tank S15	H ₂ SO ₄ – 50%	-33.926	25.527	1.8	2	2	45
S16	Tank S16	H ₂ SO ₄ – 50%	-33.926	25.527	1.8	2	2	45
E1	Tank E1	Holding water (Not in use)	-33.926	25.573	6	14	11	350
E2	Tank E2	Mixing pH Neutral	-33.926	25.573	6	14	11	350
E3	Tank E3	Holding water (Not in use)	-33.926	25.573	6	14	11	350
E4	Tank E4	Mixing pH Neutral	-33.926	25.573	6	14	11	350

Unique Area Source ID	Source Name	Source Description	Latitude (decimal degrees) of SW corner	Longitude (decimal degrees) of SW corner	Height of Release Above Ground (m)	Length of Area (m)	Width of Area (m)	Angle of Rotation from True North (°)
E5	Tank E5	Caustic -25%	-33.926	25.573	6	14	11	350
E6	Tank E6	Caustic - 25%	-33.926	25.573	6	14	11	350
E7	Tank E7	Flocculant (Not in use)	-33.926	25.573	6	14	11	350
RC1	Recovery Bath 1	Acid Recovery for Re-Use 35%	-33.926	25.573	1.2	8	11	260
RC2	Recovery Bath 2	Acid Recovery for Re-Use 35%	-33.926	25.573	1.2	8	6	260

7.4.6 Area and/or Line Source Emissions

Attach a copy of all air pollution monitoring reports conducted for the last three years.

As per 5.4.5 ID	Pollutant Name	Maximum Daily Release Rate (tons/day)	Average Annual Release Rate (tons/annum)	Emission Hours (e.g. 07h00 – 17h00)	Type of Emission (Continuous / Intermittent)	Wind Dependent (Yes / No)

7.4.7 Area and/or Line Source – Management and Mitigation Measures

Provide information on proposed management and mitigation measures as well as those as set out in currently held atmospheric emission licence(s), where applicable, for existing operations at the site of the works.

A fugitive emissions management plan must be included for listed activities that are likely to generate such emissions.

As per 5.4.5 ID	Description of Specific Measures	Compliance to Specific Measures (Yes/No)	Timeframe for Implementation of Specific Measures	Method of Monitoring Measure Effectiveness	Contingency Measure
S1 to S12 E1 to E7 RC1 – RC2	New acid storage facility built with higher bunded walls and fibre glass flooring.	Yes	August 2021	Daily Inspection	Signage and Training
	Second acid storage area to be completed and fixed with higher bunded walls and fibre glass flooring.	Yes	November 2021	Daily Inspection	Signage and Training
	Existing acid storage area to be refurbished in line with current standards.	Yes	December 2021	Daily Inspection	Signage and Training

7.4.8 Area and/or Line Source Emission Estimation Information

As per 5.4.5 ID	Basis for Emission Rates

7.4.9 Spatial Representation of Processes and Sources

Attach site layout diagram(s) (plan view and to scale, include true north arrow and scale) indicating:

- location of unit processes, plants, buildings, stacks, stockpiles and roads.
- location of point and area sources listed with unique stack ID and unique area ID, respectively, specified.

Indicate clearly on the above graphics the listed activity or activities applied for in this application. Alternatively, provide additional graphics for the listed activity or activities applied for.



8 APPLIANCES AND MEASURES TO PREVENT AIR POLLUTION

8.1 Appliances and Control Measures

Provide information on appliances and measures implemented to prevent air pollution for the entire operation at the site of the works, highlighting information for listed activity or activities proposed in respect of this application.

Provide supporting documentations that the best practicable environmental options, currently available, have been taken to prevent, control, abate or mitigate the pollution.

Appliances				Air Pollution Control Technology							
Associated Unique Stack ID	Appliance / Process Equipment Number	Appliance Type / Description	Appliance Serial Number	Appliance Manufacture Date	Product Name and Model	Technology Type	Commission Date	Date of Significant Modification / Upgrade	Design Capacity	Permitted Minimum Control Efficiency (%)	Permitted Minimum Utilization (%)
AX1	TBSWS1	TBS Assembly Wet Scrubber	14088	2012	Air & Chilled Technologies IC55	Wet Scrubber	2012	2012	25 000 m³/h	95	100
AX2	CRS01	Charge Room Extraction Stack 1 Scrubber	GE110219SA	2012	Kustan T1002H	Wet Acid Scrubber	2012	2012	65 000 m³/h	95	100
AX3	CRS02	Charge Room Extraction Stack 2 Scrubber	GE110219SA	2012	Kustan T1002H	Wet Acid Scrubber	2012	2012	70 000 m³/h	95	100
AX4	BMOB/4	BM Assembly Extraction No. 3 Bag Filter	DC3446	2025	Donaldson	Cartridge	2026	2026	25,415 m³/h	99	100

Appliances				Air Pollution Control Technology							
Associated Unique Stack ID	Appliance / Process Equipment Number	Appliance Type / Description	Appliance Serial Number	Appliance Manufacture Date	Product Name and Model	Technology Type	Commission Date	Date of Significant Modification / Upgrade	Design Capacity	Permitted Minimum Control Efficiency (%)	Permitted Minimum Utilization (%)
AX6	BMOB/1	BM Assembly Extraction No. 1 Bag Filter	DC3445	2008	Donaldson	Cartridge	2008	2008	45 000 m ³ /h	99	100
AX7	BMOB/2	BM Assembly Extraction No. 2 Bag Filter	DC3445	2008	Donaldson	Cartridge	2008	2008	45 000 m ³ /h	99	100
AX8	BMOB/3	BM Assembly Extraction No. 3 Bag Filter	DC3445	2008	Donaldson	Cartridge	2008	2008	36 000 m ³ /h	99	100
AX9	DCS1GM	Plate Making Extraction Stack Scrubber	DCS1GM	2012	Gralex Polytech	Wet Scrubber	2012	2012	75 000 m ³ /h	95	100
AX10	DCS2	Lead Oxide Manufacturing Extraction Scrubber	DCS2	2012	Air & Allied Technologies IC55	Wet Scrubber	2012	2012	25 000 m ³ /h	95	100
AX12	PMBAGSM1	Oxide Mill Silo Stack (Negative Plates) Baghouse	PMBAGSM1	2009	Eagle	Baghouse	2009	2009	5 000 m ³ /h	99	100
	PMHEP1	Silo 1-3 Hepafilter	PMHEP1	2009	Techmatic	Baghouse	2009	2009	5 000 m ³ /h	99	100
AX13	PMBAGSM1	Oxide Mill Silo Stack (Positive Plates) Baghouse	PMBAGSM1	2009	Eagle	Baghouse	2009	2010	5 000 m ³ /h	99	100

Appliances				Air Pollution Control Technology							
Associated Unique Stack ID	Appliance / Process Equipment Number	Appliance Type / Description	Appliance Serial Number	Appliance Manufacture Date	Product Name and Model	Technology Type	Commission Date	Date of Significant Modification / Upgrade	Design Capacity	Permitted Minimum Control Efficiency (%)	Permitted Minimum Utilization (%)
	PMHEP2	Silo 4-6 Hepafilter	PMHEP2	2009	Techmatic	Baghouse	2009	2009	5 000 m³/h	99	100
AX14	PMPC1	Post-Classifier Bag Filters 1,2 and 3	PMPC1	Unknown	Eagle	Baghouse	2009	2010	2 500 m³/h	99	100
AX15	PMPC2	Post-Classifier Bag Filters 4,5 and 6	PMPC2	Unknown	Eagle	Baghouse	2009	2010	2 500 m³/h	99	100
AX16	PMM1	Paste Mixer Stack Scrubber	PMM1	Unknown	Unknown	Wet Scrubber	2009	2009	3 000 m³/h	95	100
AX22	BMA1	BM Assembly Extraction Stack 4 Bag Filter	BMA1	Unknown	Donaldson	Cartridge	2009	2009	5 000 m³/h	99	100
AX23	PE1	Proposed Expansion Stack 1 Bag Filter	PE1	Unknown	Unknown	Baghouse	2024	NA	12 000 m³/h	99	100
AX24	PE2	Proposed Expansion Stack 2 Bag Filter	PE2	Unknown	Donaldson	Cartridge	2024	NA	20 000 m³/h	99	100
AX25	PE3	Proposed Expansion Stack 3 Bag Filter	PE3	Unknown	Donaldson	Cartridge	2024	NA	5 000 m³/h	99	100

Appliances				Air Pollution Control Technology							
Associated Unique Stack ID	Appliance / Process Equipment Number	Appliance Type / Description	Appliance Serial Number	Appliance Manufacture Date	Product Name and Model	Technology Type	Commission Date	Date of Significant Modification / Upgrade	Design Capacity	Permitted Minimum Control Efficiency (%)	Permitted Minimum Utilization (%)
AX26	PE4	Proposed Expansion Stack 4	PE4	Unknown	Donaldson	Cartridge	2024	NA	5 000 m ³ /h	99	100
AX27	PE5	Proposed Expansion Stack 5 Bag Filter	PE5	Unknown	Donaldson	Cartridge	2024	NA	5 000 m ³ /h	99	100

8.2 Start Up, Maintenance and Shut Down Conditions

List potential start up, maintenance, shut down, upset conditions and associated responses related to the operations at the site of the works, highlight possible releases and responses for the proposed listed activity or activities in respect of the current application.

Unit Process	Description of Occurrence of Potential Releases	Pollutant(s) Released	Briefly Outline Back Up Plan
Receiving	N/A	N/A	N/A
Storage	N/A	N/A	N/A
Lead Oxide Production	Abatement equipment	Particulate Matter, Lead	Cease operations
Plate Production	Stacks are served by abatement equipment	Particulate Matter, Lead	Cease Operations
Battery Assembly	Stacks are served by abatement equipment	SO ₂ , SO ₃ , NO _x	Cease Operations
Charging	Stacks are served by abatement equipment	SO ₂ , SO ₃ , NO _x	Cease Operations
Cleaning	N/A	N/A	N/A
Shipping	N/A	N/A	N/A

8.3 Environmental Management System

Atmospheric Emission Licence holders are required to establish an Environmental Management System (EMS) that gives effect to the principle of continuous improvement. The EMS must as a minimum provide for the actions listed below. Specify dates by which the following actions have been / will be taken for the entire operation at the site of the works:

Item	Action	Date Completed / Due Date
1	Identify and quantify potential for environmental impacts	2004
2	Prioritise the identified impacts	2004
3	Identify appropriate preventative and corrective actions	2004
4	Develop responsive management controls, systems and procedures	2004
5	Identify improvement projects to be added to the five-year Environmental Improvement Programme	11 October 2023

Attach a copy of the Environmental Management System and most recent ISO certification audit report, where applicable



American Systems Registrar, LLC, a provider of third-party system registration and accredited by the ANSI National Accreditation Board attests that:

AUTO X (PTY) LTD

1 CHR. LINDSAY & STANFORD ROAD, NEAVE TOWNSHIP
PORT ELIZABETH 6000 SOUTH AFRICA



with a scope of:

MANUFACTURE OF LEAD ACID BATTERIES

has established a quality management system that is in conformance with the International Quality System Standard

ISO 9001:2015

ASR Certificate Number:	5785
Date of Certification:	July 27, 2021
Date of Certification Expiration:	July 26, 2024
Date of Initial Registration:	August 29, 2014
Reissues:	
Re-issue Date:	

Richelle Kingie
President

CERTIFICATE OF REGISTRATION

8.4 Energy Conservation Measures

List activities to improve energy utilisation and efficiency which are currently implemented at the site of works, highlighting proposed measures to be implemented in respect of the proposed listed activity or activities.

bsi.  

Certificate of Registration

OCCUPATIONAL HEALTH & SAFETY MANAGEMENT SYSTEM - ISO 45001:2018

This is to certify that: **AUTOX (PTY) LTD**
Corner of Lindsay & Stanford Road
Neave Township
6000 Gqeberha
South Africa

Holds Certificate No: **OHS 837043**
and operates an Occupational Health and Safety Management System which complies with the requirements of ISO 45001:2018 for the following scope:

Manufacture of Lead Acid Batteries

For and on behalf of BSI: 
Matt Page, Senior Vice President, EMEA Assurance

Original Registration Date: 2015-02-15
Latest Revision Date: 2026-02-05
Effective Date: 2026-02-05
Expiry Date: 2027-02-11

Page: 1 of 1

...making excellence a habit.™

This certificate was issued electronically and remains the property of BSI and is bound by the conditions of contract.
An electronic certificate can be authenticated [online](https://www.bsigroup.com/ClientDirectory).
Printed copies can be validated at www.bsigroup.com/ClientDirectory
Information and Contact: BSI, Kitemark Court, Davy Avenue, Knowlton, Milton Keynes MK9 6PP. Tel: +44 (0)1908 9000

bsi.  

Certificate of Registration

ENVIRONMENTAL MANAGEMENT SYSTEM - ISO 14001:2015

This is to certify that: **AUTOX (PTY) LTD**
Corner of Lindsay & Stanford Road
Neave Township
6000 Gqeberha
South Africa

Holds Certificate No: **EMS 837042**
and operates an Environmental Management System which complies with the requirements of ISO 14001:2015 for the following scope:

Despatch and distribution of batteries as well as renewable energy products

For and on behalf of BSI: 
Matt Page, Senior Vice President, EMEA Assurance

Original Registration Date: 2025-02-15
Latest Revision Date: 2026-02-05
Effective Date: 2026-02-05
Expiry Date: 2027-02-11

Page: 1 of 1

...making excellence a habit.™

This certificate was issued electronically and remains the property of BSI and is bound by the conditions of contract.
An electronic certificate can be authenticated [online](https://www.bsigroup.com/ClientDirectory).
Printed copies can be validated at www.bsigroup.com/ClientDirectory
Information and Contact: BSI, Kitemark Court, Davy Avenue, Knowlton, Milton Keynes MK9 6PP. Tel: +44 (0)1908 9000

Energy Conservation Measure	Date Implemented / to be Implemented	Target	Date by which to Achieve Target	Progress Monitoring & Reporting Method
LED Lighting	-	To reduce the facility's energy consumption.	-	-
ISO 50001 Certification	2023	Certification received for energy management.	2023	-
Insulation of Chilled Water Lines	December 2022	Insulation of the chilled water lines aids in keeping the water cool, thus requiring less energy from the cooling towers.	-	-
Purchase of More Efficient Generators	2023	The increased efficiency of new generators will reduce the fuel consumption of generators and the emissions that are emitted into the atmosphere.	-	-
Installation of Solar Panels	-	Installation of solar panels will further reduce the facility's electricity demand.	-	-
Replacement of Geysers with Solar Heaters	2023	The switch to solar heaters will reduce the facility's electricity demand.	-	-

8.5 Cleaner Production Targets

List cleaner production measures which has been or are currently being implemented at the site of works, highlighting proposed measures to be implemented in respect of the proposed listed activity or activities.

Cleaner Production Measure	Date Implemented / to be Implemented	Target	Date by which to Achieve Target	Progress Monitoring & Reporting Method
Rainwater harvesting plant	Phase 1 completed – 31 March 2023;	Rainwater is collected to be used in production areas at AutoX.	End of 2025	Phase 1 – Completed on the 31 st of March 2023. Phase 2 – To be completed at the end of 2024. Phase 3 – To be completed at the end of 2025.
Reverse Osmosis (RO) Plant	2023	Improving water quality for processes.	November 2023	Phase 1 -Completed on the 31 st of March 2023. Phase 2 – To be completed by November 2023.
Improved Stack Sampling Accuracy	2023	Improve accuracy of stack sampling.	2023	-
Identification of Unregulated Stacks	2023	Identify critical samples not present in the current AEL.	2023	-
Tracking of Absolute Emissions	2023	Target reductions and used as a tool to measure annual reductions.	2023	Ongoing
Air Dispersion Modelling Investigation	2023	To identify modifications that can be made to reduce ambient particulate matter and lead concentrations.	2023	Atmospheric Impact Report – Completed on the 14 th of September 2023.
Repairs to Ducting of AX9 Stack	2023	Repair the ducting on AX9.	2023	Complete
Cleaning and Maintenance of Abatement Equipment	2023	Regular maintenance of abatement equipment to ensure that the equipment works well to reduce the concentration of emissions.	Conducted regularly	-
Audit on Ducting	2021	Assessment of ducting to ensure flowrates are sufficient for extracting emissions from workstations.	2021	-

Asbestos Management Plan	2023	To replace all asbestos roofing with steel sheeting.	2023	-
Welding of TBS Windows	2021	Welding windows closed ensures that all of the emissions that are released are extracted through extraction systems that feed into the abatement equipment.	2021	-
Switch to Lithium Ion Fork Trucks	-	Switching to Lithium-Ion results in a reduction of Scope 1 emissions for AutoX.	-	-
Green Initiatives	-	Trees are planted based on the number of batteries that are purchased.	-	-

8.6 Routine Reporting and Record-Keeping

8.6.1 Complaints Register

Is a complaints register maintained at the site works?

	Yes	X
	No	
	To be initiated, by date:	

In the event that a complaints register is maintained, please provide a copy of complaints received and corrective actions taken over the past two years.

8.6.2 Non-compliance with Current Atmospheric Emission Licence Conditions

If atmospheric emission licence(s) are currently held, summarise instances of non-compliance with the conditions of such atmospheric emission licence(s) which have occurred over the past two years:

Unique Stack ID	Pollutant Released	Emission Limit Exceeded	Root Cause Analysis	Measures Implemented to Prevent Recurrence	Date by which Measures were / will be Implemented

9 DISPOSAL OF WASTE AND EFFLUENTS ARISING FROM AIR POLLUTION MITIGATION MEASURES

Provide the following information for any waste and effluent arising from any air pollution mitigation measures that are currently in place at the site of the works:

Unique Stack or Area ID (As per 5.4.1 or 5.4.5 above)	Waste / Effluent Type	Hazardous Components Present	Method of Disposal	Licence Status (Provisional or Final)
AX1; 2; 3; 9; 10; 16	Industrial effluent	Arsenic Boron Cadmium Cobalt Lead Molybdenum Selenium Mercury	Industrial effluent is discharged into the municipal sewerage system.	Final
AX4; 6; 7; 8; 12; 13; 14; 15; 22; 23; 24; 25; 26; 27	Dustfall	N/A	Dust is collected by Frys Metals – specializing in lead recycling operations.	Final

Provide information on any waste and effluent which will arise from air pollution mitigation measures proposed for implementation for the listed activity or activities dealt with in this application:

Unique Stack or Area ID (As per 5.4.1 or 5.4.5 above)	Waste / Effluent Type	Hazardous Components Present	Method of Disposal	Licence Status (Provisional or Final)
AX23; 24; 25; 26; 27	Dustfall	N/A	Dust is collected by Frys Metals – specializing in lead recycling operations.	Final

10 LEGAL COMPLIANCE & COMPETENCE

It is imperative that the holder of the atmospheric emission licence is a fit and proper person in terms of section 49 of the National Environmental Management: Air Quality Act 39 of 2004 for the purposes of an application in terms of chapter 5 of the said act. To assess the applicant's legal compliance and competence, please disclose the following:

10.1 Legal compliance

Question	Yes/No	Details (If Yes)
Has the applicant ever been found guilty or issued with a non compliance notice in terms of the National Environmental Management Act, the Atmospheric Pollution Prevention Act (APPA) or any other legislation applicable to air quality?	TBC	
Has the applicant ever held a provisional/registration certificate issued in terms of the APPA, provisional / atmospheric emission licence in terms of the Air Quality Act 39 of 2004 that has been refused, suspended or revoked?	TBC	
Has the applicant been a director or senior manager who is or was a director or manager of a company, a juristic person of firm to whom the above two questions apply?	TBC	
Has the applicant ever been issued with a non compliance notice or letter in terms of any South African Law?	TBC	
<i>NB: Details required any information that the applicant wants the Licensing Authority to take into consideration in determining whether they are a "fit and proper person" and this includes why the offence happened and measures in place to prevent recurrence.</i>		

10.2 Technical Competence

--	--

Will an emission control officer be appointed? If "yes" state the name.	Yes. Tshepo Nkolanyane
If "yes", please state the qualifications and experience	5+ years of experience Qualifications: MSc in Environmental Science
If "yes" please state the duties and responsibilities	Regulatory compliance Monitoring and reporting emissions Implementing emission reduction strategies Environmental audits and risk assessments Continuous improvement and training

PART C

Declaration of accuracy of information provided:

Application for an atmospheric emission licence / provisional atmospheric emission licence as envisaged in chapter 5 of the National Environmental Management: Air Quality Act, 2004 (Act No. 39 of 2004).

I, Tshepo Nkolanyane, declare that the information provided in this application or attached to the application is, to the best of my knowledge, in all respects factually true and correct. I am aware that the supply of false or misleading information in the application form is a criminal offence in terms of section 51(1)(f) of the Act.

Signed at AutoX on this day 16 April of 2026



SIGNATURE

Environmental Specialist

CAPACITY OF SIGNATORY